

Name: _____ Class: _____

Study Guide for Q3 End of Quarter Test

Topics:

- Cellular Structure
- Characteristics of Organisms
- Ecosystems, Adaptations, Food Webs
- Human Body and Organ Systems

I can...

- ☐ explain the main components of the cell theory
- ☐ identify common parts of a cell and explain their primary functions.
- ☐ differentiate between prokaryotic and eukaryotic cells, as well as animal and plant cells
- ☐ describe, in detail, the defining characteristics of animals and plants.
- ☐ classify plants using the terms: vascular, non-vascular, bryophyte, gymnosperm, angiosperm
- ☐ identify basic structures in plant anatomy, including root and shoot systems, vascular tissue, and seed-producing bodies.
- ☐ label the reproductive structures of a flowering plant.
- ☐ group organisms, including plants, by similar characteristics.
- ☐ give examples of the interactions of living and non-living things in an environment.
- ☐ illustrate the movement of energy in an ecosystem via food chains and food webs.
- ☐ give examples of organisms from different energy roles in an ecosystem
- ☐ illustrate the interdependence of species within an ecosystem.
- ☐ compare common characteristics of organisms in biomes.
- ☐ Illustrate the advantages of biodiversity.
- ☐ explain the basics of the individual systems and how they work together to perform a vital function.
- ☐ discuss the relationships between structures and functions of major body systems (cardiovascular, skeletal, respiratory, digestive, nervous, integumentary, and muscular) and organs.
- ☐ classify human body functions as voluntary and involuntary responses

Use this list to see which skills you feel confident about, and which skills you want to review and practice.
Use the questions in this packet to test your understanding.

Resources:

- class notes
- quizzes
- homework books
- student hour

Cells

In the space below, **draw** a **plant cell** and an **animal cell**. Include as many **organelles** as you can.

Plant Cell	Animal Cell

Draw and explain the difference between **prokaryotic** cells and **eukaryotic** cells. Include examples of organisms from each category.

Prokaryotic	Eukaryotic
Diagram:	Diagram:
Definition:	Definition:
Examples:	Examples:

What are the 3 main principals of **cell theory**?

1.) _____

2.) _____

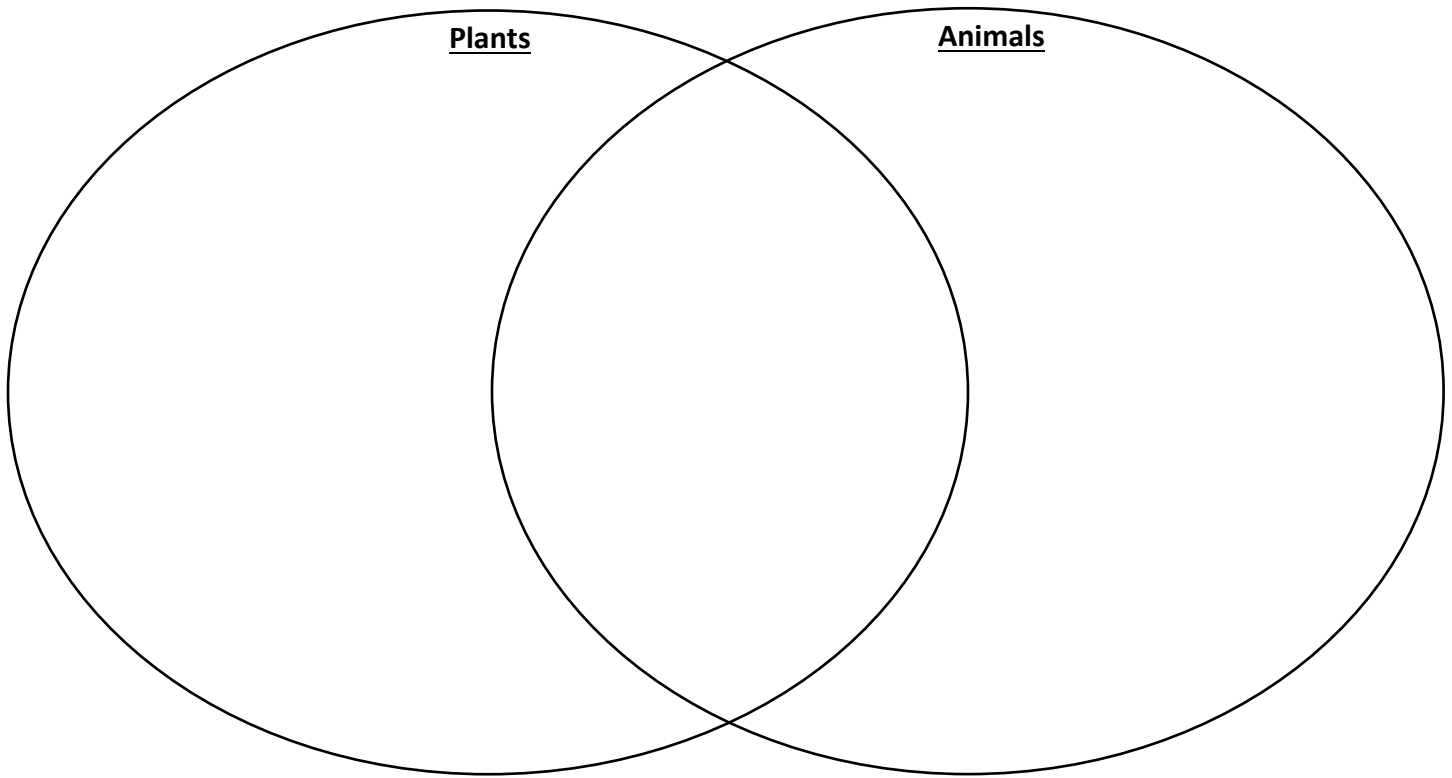
3.) _____

Find the **cell structure** that matches each definition below:

Cell Structure	Definition
	Controls all cell functions and contains DNA
	Controls what goes in and out of the cell
	The jelly-like substance that all the organelles float in
	The transportation network of the cell. May be covered in ribosomes to make and fold proteins
	Receives substances from the Endoplasmic Reticulum (ER) and delivers them
	Use glucose to make energy molecules called ATP
	Breaks down large chunks of food into usable sizes in animal cells
	Uses energy from sunlight to make glucose and oxygen from water and carbon dioxide
	Storage area, mainly of water in plant cells
	Rigid material that surrounds the cell membrane that gives the plant its shape
	makes protiens from amino acids using mRNA instructions
	large molecules needed for life. They contain carbon, oxygen, nitrogen and hydrogen
	building blocks of proteins that come in different shapes

Characteristics of Organisms

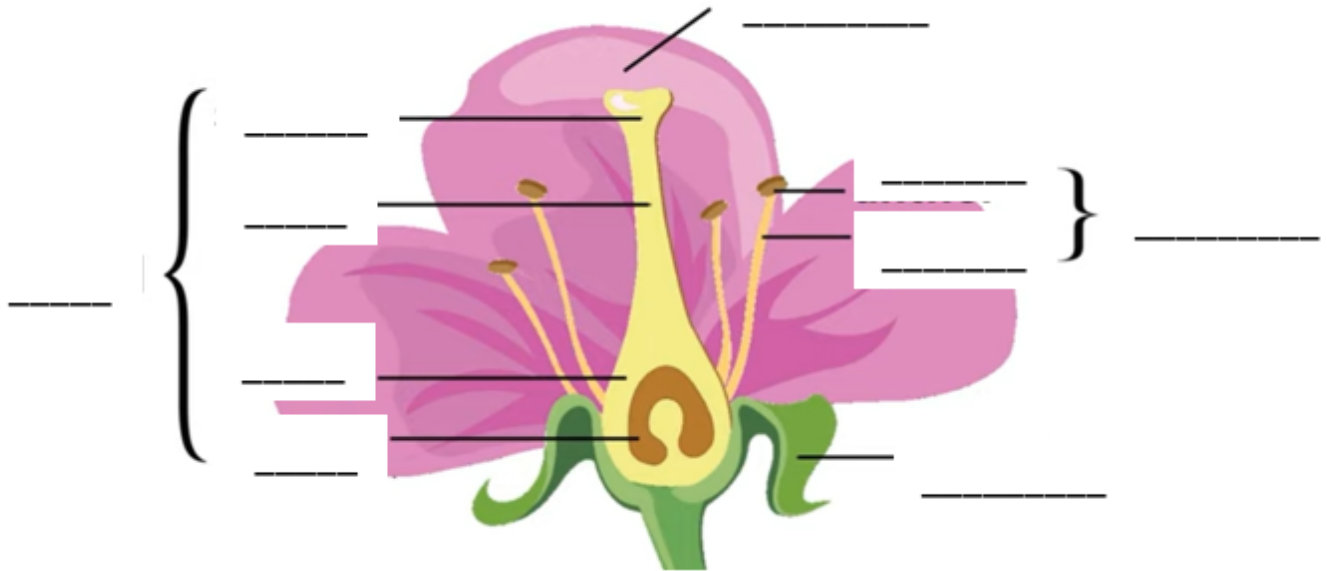
Complete the Venn diagram with common **characteristics of plants and animals**.



In the space below, create a **dichotomous key** for the different kinds of plants.

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Complete the diagram of a flower.



Ecosystems:

In the space below, draw and write to make a model of a food web. Be sure to label the following:

- ☐ primary consumer
- ☐ apex predator

- ☐ secondary consumer
- ☐ primary producer

- ☐ decomposer
- ☐ prey

Blank space for drawing and writing a model of a food web.

1.) Choose one species to remove from your food web (do not erase it). What will change as a result of that species missing?

2.) How could you make your ecosystem safer from disruptions like invasive species, new diseases, and competition from humans?

3.) What **abiotic factors** make life possible for the organisms in the ecosystem you drew above?

4.) Choose one animal in your ecosystem model and describe **four adaptations** that helps it survive.
Choose **2 physical adaptations** and **2 behavioral adaptations**.

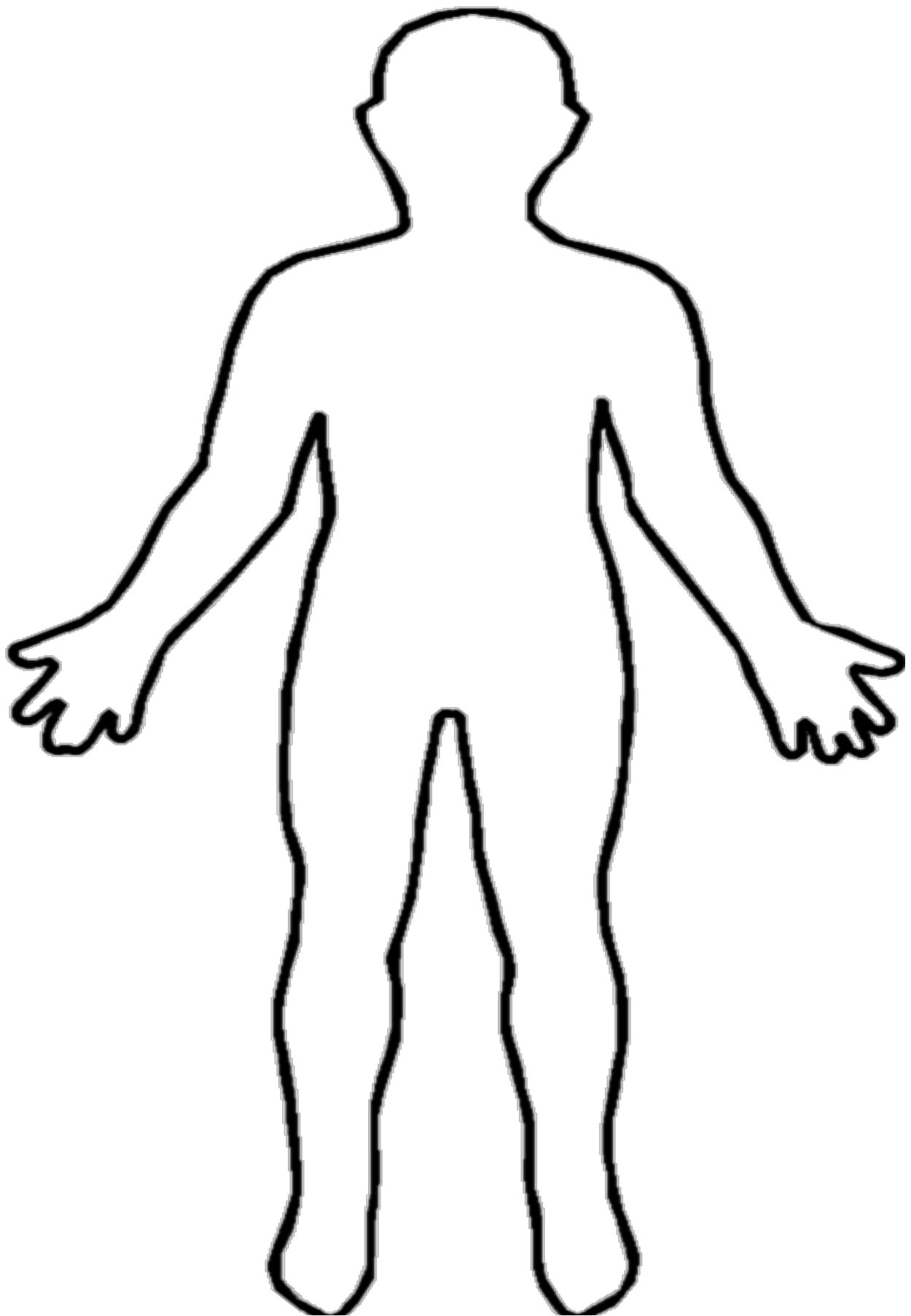
Animal: _____	Adaptation	How it helps the animal survive
Physical Adaptations		
Behavioral Adaptations		

Organ Systems:

In the space below, model any 3 or the **organ systems** from the following table.

My Model Shows:	Organ systems		
1.)	skeletal	circulatory	digestive
2.)	muscular	respiratory	nervous
3.)		integumentary	

Label the **main parts** of each system and show where they overlap and connect.



4.) Explain how the **functions** of each system you chose **support each other**. What does each organ system need from the others to perform its function?

Chosen Systems	What it needs from System 1	What it needs from System 2	What it needs from System 3
System 1:			
System 2:			
System 3:			

5.) Which of the body's functions are involuntary? List as many as you can.
